Australia can be affected by both warm fronts and cold fronts, however cold fronts are more common and have a greater impact on the Australian region. A cold front is formed when cold dense air advances equatorwards, causing warm air to be forced aloft over its sloping surface. A warm front is formed when warm air of lower density moves polewards, sliding over the sloping surface formed by a colder air mass.

Stationary fronts are simply fronts which remain very slow-moving or near stationary. Another type of front is the occluded front. Both of these are much less common in the Australian region.

Frontal systems affect all of southern Australia. They can occur at any time of year, however they have the greatest impact during the winter months. Fronts generally move across southern Australia from west to east, and can be in the Australian region from a couple of days to a week.

54. **The Madden-Julian Oscillation (MJO):**

- The Madden-Julian Oscillation (MJO) is a glob- al-scale feature of the tropical atmosphere.
- The MJO is the major fluctuation in tropical weather on weekly to monthly timescales. The MJO can be characterised as an eastward moving "pulse" of cloud and rainfall near the equator that typically recurs every 30 to 60 days. However, the signal of the MJO in the tropical atmosphere is not always present.
- MJO effects are most evident over the Indian Ocean and western equatorial Pacific. It influences the timing, development and strength of the major global monsoon patterns, including the Indian and Australian monsoons.
- Tropical cyclones are also more likely to develop in association with certain phases of a strong MJO event.
- The MJO is associated with variations in wind, cloudiness, and rainfall. Most tropical rainfall comes from tall thunderstorms which have very cold tops. Thunderstorms that have cold tops emit only low levels of longwave radiation. Therefore, the MJO can be monitored by using satellite measurements of outgoing longwave radiation (OLR) to identify areas of cloudiness (low OLR) within the tropics.